NIFTY 50 Regime Change Pipeline

Run the project in order: Phase 1 builds a local NIFTY 50 snapshot and causal features; Phase 2 fits and runs frozen PELT–L2 and PELT–RBF detectors on validation; Phase 3 compares validation labels with a training-fit, forward-filtered three-state Gaussian HMM. The final held-out PELT test cell is present but gated off by default.

Look-ahead controls: Features for date *t* use closes only through the previous observed session. Splits are chronological. Phase 2 fits scales, RBF bandwidth, and descriptive-volatility cutoffs on training only. The HMM is also fit on training only and uses forward filtering on validation. Do not tune settings on test. Keep each output directory new and empty.

0. Install dependencies

Run once in the active Jupyter environment. Restart the kernel if Jupyter asks you to.

```
%pip install -r requirements.txt
```

```
^C
Note: you may need to restart the kernel to use updated packages.

ERROR: Could not install packages due to an OSError: [WinError 32] The
process cannot access the file because it is being used by another
process: 'C:\\Users\\pqrth\\OneDrive\\Desktop\\RegimeChanges-main\\
RegimeChanges-main\\.venv\\Lib\\site-packages\\pandas\\tests\\series\\
methods\\test_searchsorted.py'
Check the permissions.


Requirement already satisfied: numpy>=1.21 in .\.venv\Lib\site-
packages (from -r requirements.txt (line 1)) (2.4.6)
Collecting pandas>=1.3 (from -r requirements.txt (line 2))
  Using cached pandas-3.1.0-cp311-cp311-win_amd64.whl.metadata (18 kB)
Collecting matplotlib>=3.5 (from -r requirements.txt (line 3))
  Using cached matplotlib-3.11.2-cp311-cp311-win_amd64.whl.metadata
(80 kB)
Collecting yfinance>=0.2 (from -r requirements.txt (line 4))
  Using cached yfinance-1.7.0-py3-none-any.whl.metadata (6.7 kB)
Collecting scipy>=1.7 (from -r requirements.txt (line 5))
  Using cached scipy-1.17.1-cp311-cp311-win_amd64.whl.metadata (60 kB)
Collecting ruptures>=1.1.10 (from -r requirements.txt (line 6))
  Using cached ruptures-1.1.10-cp311-cp311-win_amd64.whl.metadata (9.4
kB)
Requirement already satisfied: tzdata>=2022.1 in .\.venv\Lib\site-
packages (from -r requirements.txt (line 7)) (2026.6)
Collecting hmmlearn>=0.3.3 (from -r requirements.txt (line 8))
  Using cached hmmlearn-0.3.3-cp311-cp311-win_amd64.whl.metadata (3.1
kB)
```

```
Requirement already satisfied: python-dateutil>=2.9.0 in .\.venv\Lib\
site-packages (from pandas>=1.3->-r requirements.txt (line 2)) (3.0.0)
Collecting contourpy>=1.0.1 (from matplotlib>=3.5->-r requirements.txt
(line 3))
  Using cached contourpy-1.3.3-cp311-cp311-win_amd64.whl.metadata (5.5
kB)
Collecting cycler>=0.10 (from matplotlib>=3.5->-r requirements.txt
(line 3))
  Using cached cycler-0.12.1-py3-none-any.whl.metadata (3.8 kB)
Collecting fonttools>=4.28.2 (from matplotlib>=3.5->-r
requirements.txt (line 3))
  Using cached fonttools-4.68.0-cp311-cp311-win_amd64.whl.metadata
(142 kB)
Collecting kiwisolver>=1.3.1 (from matplotlib>=3.5->-r
requirements.txt (line 3))
  Using cached kiwisolver-1.5.1-cp311-cp311-win_amd64.whl.metadata
(5.2 kB)
Requirement already satisfied: packaging>=20.0 in .\.venv\Lib\site-
packages (from matplotlib>=3.5->-r requirements.txt (line 3)) (26.3)
Requirement already satisfied: pillow>=9 in .\.venv\Lib\site-packages
(from matplotlib>=3.5->-r requirements.txt (line 3)) (13.0.0)
Requirement already satisfied: pyparsing>=3 in .\.venv\Lib\site-
packages (from matplotlib>=3.5->-r requirements.txt (line 3)) (3.3.3)
Requirement already satisfied: beautifulsoup4>=4.11.1 in .\.venv\Lib\
site-packages (from yfinance>=0.2->-r requirements.txt (line 4))
(4.15.0)
Collecting curl_cffi>=0.15 (from yfinance>=0.2->-r requirements.txt
(line 4))
  Using cached curl_cffi-0.17.0-cp310-abi3-win_amd64.whl.metadata (19
kB)
Collecting lxml>=4.9.0 (from yfinance>=0.2->-r requirements.txt (line
4))
  Using cached lxml-6.1.3-cp311-cp311-win_amd64.whl.metadata (3.4 kB)
Requirement already satisfied: multitasking>=0.0.7 in .\.venv\Lib\
site-packages (from yfinance>=0.2->-r requirements.txt (line 4))
(0.0.13)
Requirement already satisfied: peewee>=3.16.2 in .\.venv\Lib\site-
packages (from yfinance>=0.2->-r requirements.txt (line 4)) (4.5.3)
Requirement already satisfied: platformdirs>=2.0.0 in .\.venv\Lib\
site-packages (from yfinance>=0.2->-r requirements.txt (line 4))
(4.13.7)
Requirement already satisfied: protobuf>=3.19.0 in .\.venv\Lib\site-
packages (from yfinance>=0.2->-r requirements.txt (line 4)) (7.36.2)
Requirement already satisfied: pytz>=2022.5 in .\.venv\Lib\site-
packages (from yfinance>=0.2->-r requirements.txt (line 4)) (2026.5)
Requirement already satisfied: requests>=2.31 in .\.venv\Lib\site-
packages (from yfinance>=0.2->-r requirements.txt (line 4)) (2.34.2)
Requirement already satisfied: websockets>=13.0 in .\.venv\Lib\site-
packages (from yfinance>=0.2->-r requirements.txt (line 4)) (17.3)
```

```
Collecting scikit-learn!=0.22.0,>=0.16 (from hmmlearn>=0.3.3->-r
requirements.txt (line 8))
  Using cached scikit_learn-1.9.1-cp311-cp311-win_amd64.whl.metadata
(9.3 kB)
Requirement already satisfied: soupsieve>=1.6.1 in .\.venv\Lib\site-
packages (from beautifulsoup4>=4.11.1->yfinance>=0.2->-r
requirements.txt (line 4)) (3.0.4)
Requirement already satisfied: typing-extensions>=4.0.0 in .\.venv\
Lib\site-packages (from beautifulsoup4>=4.11.1->yfinance>=0.2->-r
requirements.txt (line 4)) (4.16.0)
Requirement already satisfied: cffi>=2.1.1 in .\.venv\Lib\site-
packages (from curl_cffi>=0.15->yfinance>=0.2->-r requirements.txt
(line 4)) (2.1.1)
Requirement already satisfied: certifi>=2026.10.19 in .\.venv\Lib\
site-packages (from curl_cffi>=0.15->yfinance>=0.2->-r
requirements.txt (line 4)) (2026.10.23)
Requirement already satisfied: pycparser in .\.venv\Lib\site-packages
(from cffi>=2.1.1->curl_cffi>=0.15->yfinance>=0.2->-r requirements.txt
(line 4)) (3.12)
Requirement already satisfied: six>=1.6 in .\.venv\Lib\site-packages
(from python-dateutil>=2.9.0->pandas>=1.3->-r requirements.txt (line
2)) (1.17.0)
Requirement already satisfied: charset_normalizer<4,>=2 in .\.venv\
Lib\site-packages (from requests>=2.31->yfinance>=0.2->-r
requirements.txt (line 4)) (3.5.2)
Requirement already satisfied: idna<4,>=2.5 in .\.venv\Lib\site-
packages (from requests>=2.31->yfinance>=0.2->-r requirements.txt
(line 4)) (3.20)
Requirement already satisfied: urllib3<3,>=1.26 in .\.venv\Lib\site-
packages (from requests>=2.31->yfinance>=0.2->-r requirements.txt
(line 4)) (2.8.0)
Collecting joblib>=1.4.0 (from scikit-learn!=0.22.0,>=0.16-
>hmmlearn>=0.3.3->-r requirements.txt (line 8))
  Using cached joblib-1.6.0-py3-none-any.whl.metadata (6.5 kB)
Collecting narwhals>=2.0.1 (from scikit-learn!=0.22.0,>=0.16-
>hmmlearn>=0.3.3->-r requirements.txt (line 8))
  Using cached narwhals-2.28.0-py3-none-any.whl.metadata (15 kB)
Requirement already satisfied: threadpoolctl>=3.5.0 in .\.venv\Lib\
site-packages (from scikit-learn!=0.22.0,>=0.16->hmmlearn>=0.3.3->-r
requirements.txt (line 8)) (3.7.0)
Collecting cloudpickle>=3.0 (from joblib>=1.4.0->scikit-learn!
=0.22.0,>=0.16->hmmlearn>=0.3.3->-r requirements.txt (line 8))
  Using cached cloudpickle-3.1.2-py3-none-any.whl.metadata (7.1 kB)
Using cached pandas-3.1.0-cp311-cp311-win_amd64.whl (11.0 MB)
Using cached matplotlib-3.11.2-cp311-cp311-win_amd64.whl (9.3 MB)
Using cached yfinance-1.7.0-py3-none-any.whl (149 kB)
Using cached scipy-1.17.1-cp311-cp311-win_amd64.whl (36.6 MB)
Using cached ruptures-1.1.10-cp311-cp311-win_amd64.whl (475 kB)
Using cached hmmlearn-0.3.3-cp311-cp311-win_amd64.whl (127 kB)
```

```
Using cached contourpy-1.3.3-cp311-cp311-win_amd64.whl (225 kB)
Using cached curl_cffi-0.17.0-cp310-abi3-win_amd64.whl (2.1 MB)
Using cached cycler-0.12.1-py3-none-any.whl (8.3 kB)
Using cached fonttools-4.68.0-cp311-cp311-win_amd64.whl (1.7 MB)
Using cached kiwisolver-1.5.1-cp311-cp311-win_amd64.whl (70 kB)
Using cached lxml-6.1.3-cp311-cp311-win_amd64.whl (4.0 MB)
Using cached scikit_learn-1.9.1-cp311-cp311-win_amd64.whl (8.3 MB)
Using cached joblib-1.6.0-py3-none-any.whl (306 kB)
Using cached cloudpickle-3.1.2-py3-none-any.whl (22 kB)
Using cached narwhals-2.28.0-py3-none-any.whl (490 kB)
Installing collected packages: scipy, narwhals, lxml, kiwisolver,
fonttools, cycler, contourpy, cloudpickle, ruptures, pandas,
matplotlib, joblib, curl_cffi, yfinance, scikit-learn, hmmlearn

   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
```

```
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
```

```
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
```

```
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
```

```
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
```

```
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
```

```
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
   ---------------------------------------- 0/16 [scipy]
```

```
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
```

```
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   ----------------------------------------  0/16 [scipy]
   -- -------------------------------------  1/16 [narwhals]
   -- -------------------------------------  1/16 [narwhals]
   -- -------------------------------------  1/16 [narwhals]
   -- -------------------------------------  1/16 [narwhals]
   -- -------------------------------------  1/16 [narwhals]
   -- -------------------------------------  1/16 [narwhals]
   -- -------------------------------------  1/16 [narwhals]
   -- -------------------------------------  1/16 [narwhals]
   -- -------------------------------------  1/16 [narwhals]
   -- -------------------------------------  1/16 [narwhals]
   -- -------------------------------------  1/16 [narwhals]
   -- -------------------------------------  1/16 [narwhals]
   -- -------------------------------------  1/16 [narwhals]
   -- -------------------------------------  1/16 [narwhals]
   -- -------------------------------------  1/16 [narwhals]
   -- -------------------------------------  1/16 [narwhals]
   -- -------------------------------------  1/16 [narwhals]
   -- -------------------------------------  1/16 [narwhals]
   -- -------------------------------------  1/16 [narwhals]
   -- -------------------------------------  1/16 [narwhals]
   -- -------------------------------------  1/16 [narwhals]
   -- -------------------------------------  1/16 [narwhals]
   -- -------------------------------------  1/16 [narwhals]
   -- -------------------------------------  1/16 [narwhals]
   -- -------------------------------------  1/16 [narwhals]
   -- -------------------------------------  1/16 [narwhals]
   -- -------------------------------------  1/16 [narwhals]
   -- -------------------------------------  1/16 [narwhals]
   -- -------------------------------------  1/16 [narwhals]
   -- -------------------------------------  1/16 [narwhals]
   -- -------------------------------------  1/16 [narwhals]
   -- -------------------------------------  1/16 [narwhals]
   -- -------------------------------------  1/16 [narwhals]
   -- -------------------------------------  1/16 [narwhals]
   -- -------------------------------------  1/16 [narwhals]
   -- -------------------------------------  1/16 [narwhals]
   -- -------------------------------------  1/16 [narwhals]
   -- -------------------------------------  1/16 [narwhals]
   -- -------------------------------------  1/16 [narwhals]
   -- -------------------------------------  1/16 [narwhals]
   -- -------------------------------------  1/16 [narwhals]
   -- -------------------------------------  1/16 [narwhals]
   -- -------------------------------------  1/16 [narwhals]
   -- -------------------------------------  1/16 [narwhals]
   -- -------------------------------------  1/16 [narwhals]
```

```
  -- ------------------------------------  1/16 [narwhals]
  -- ------------------------------------  1/16 [narwhals]
  -- ------------------------------------  1/16 [narwhals]
  -- ------------------------------------  1/16 [narwhals]
  -- ------------------------------------  1/16 [narwhals]
  -- ------------------------------------  1/16 [narwhals]
  -- ------------------------------------  1/16 [narwhals]
  -- ------------------------------------  1/16 [narwhals]
  -- ------------------------------------  1/16 [narwhals]
  -- ------------------------------------  1/16 [narwhals]
  -- ------------------------------------  1/16 [narwhals]
  -- ------------------------------------  1/16 [narwhals]
  -- ------------------------------------  1/16 [narwhals]
  -- ------------------------------------  1/16 [narwhals]
  ----- -----------------------------------  2/16 [lxml]
  ----- -----------------------------------  2/16 [lxml]
  ----- -----------------------------------  2/16 [lxml]
  ----- -----------------------------------  2/16 [lxml]
  ----- -----------------------------------  2/16 [lxml]
  ----- -----------------------------------  2/16 [lxml]
  ----- -----------------------------------  2/16 [lxml]
  ----- -----------------------------------  2/16 [lxml]
  ----- -----------------------------------  2/16 [lxml]
  ----- -----------------------------------  2/16 [lxml]
  ----- -----------------------------------  2/16 [lxml]
  ----- -----------------------------------  2/16 [lxml]
  ----- -----------------------------------  2/16 [lxml]
  ----- -----------------------------------  2/16 [lxml]
  ----- -----------------------------------  2/16 [lxml]
  ----- -----------------------------------  2/16 [lxml]
  ----- -----------------------------------  2/16 [lxml]
  ----- -----------------------------------  2/16 [lxml]
  ----- -----------------------------------  2/16 [lxml]
  ----- -----------------------------------  2/16 [lxml]
  ----- -----------------------------------  2/16 [lxml]
  ------- ---------------------------------  3/16 [kiwisolver]
  ------- ---------------------------------  3/16 [kiwisolver]
  ---------- -------------------------------  4/16 [fonttools]
  ---------- -------------------------------  4/16 [fonttools]
  ---------- -------------------------------  4/16 [fonttools]
  ---------- -------------------------------  4/16 [fonttools]
  ---------- -------------------------------  4/16 [fonttools]
  ---------- -------------------------------  4/16 [fonttools]
  ---------- -------------------------------  4/16 [fonttools]
  ---------- -------------------------------  4/16 [fonttools]
  ---------- -------------------------------  4/16 [fonttools]
```

```
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
```

```
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
   ---------- ------------------------------  4/16 [fonttools]
```

```
   -------------- ------------------------  6/16 [contourpy]
   -------------- ------------------------  6/16 [contourpy]
   -------------------- --------------------  8/16 [ruptures]
   -------------------- --------------------  8/16 [ruptures]
   ---------------------- ----------------  9/16 [pandas]
   ---------------------- ----------------  9/16 [pandas]
   ---------------------- ----------------  9/16 [pandas]
   ---------------------- ----------------  9/16 [pandas]
   ---------------------- ----------------  9/16 [pandas]
   ---------------------- ----------------  9/16 [pandas]
   ---------------------- ----------------  9/16 [pandas]
   ---------------------- ----------------  9/16 [pandas]
   ---------------------- ----------------  9/16 [pandas]
   ---------------------- ----------------  9/16 [pandas]
   ---------------------- ----------------  9/16 [pandas]
   ---------------------- ----------------  9/16 [pandas]
   ---------------------- ----------------  9/16 [pandas]
```

1. Set paths and data source

Keep `phase1_nifty.py`, `phase2_cpd.py`, `phase3_hmm.py`, `requirements.txt`, and this notebook in the same project folder. Choose `DATA_SOURCE = "yahoo"` to download NIFTY 50 through yfinance, or set `DATA_SOURCE = "local_csv"` and provide a folder containing `NIFTY50.csv` with `Date,Close` columns. The CSV needs enough history for the chronological training, validation, and test periods.

Each run needs new output folders. Phase 3 uses the exact frozen PELT configuration (252/20, penalties 10/5). The final test cell is gated off so ordinary notebook runs leave the holdout closed.

```
from pathlib import Path
import json
import pandas as pd
import matplotlib.pyplot as plt

PROJECT_DIR = Path.cwd()  # change this if the notebook is not in the
project folder
PHASE1_OUTPUT = PROJECT_DIR / 'phase1_nifty50'
PHASE2_OUTPUT = PROJECT_DIR / 'phase2_nifty50_validation'
PHASE3_HMM_OUTPUT = PROJECT_DIR / 'phase3_hmm_validation'
FINAL_OUTPUT = PROJECT_DIR / 'phase2_nifty50_final_test'
DATA_SOURCE = 'yahoo'  # choose 'yahoo' or 'local_csv'
LOCAL_CSV_DIR = PROJECT_DIR / 'index_csvs'  # used only for local_csv
RUN_FINAL_TEST = True  # change to True only once, after freezing all
choices

for script in ('phase1_nifty.py', 'phase2_cpd.py', 'phase3_hmm.py'):
```

```
    assert (PROJECT_DIR / script).exists(), f'{script} must be in
PROJECT_DIR'
print('Project folder:', PROJECT_DIR)
print('Data source:', DATA_SOURCE)
print('Final test enabled:', RUN_FINAL_TEST)
```

```
Project folder: C:\Users\pqrth\OneDrive\Desktop\RegimeChanges-main\
RegimeChanges-main
Data source: yahoo
Final test enabled: True
```

2. Run Phase 1 checks

These synthetic self-tests verify the script without downloading market data.

```
import phase1_nifty
phase1_nifty.main(['--self-test'])
```

```
PASS: lag timing, trailing volatility, prefix/future invariance,
missing data, chronological splits, train-only scaling, CSV/MultiIndex
parsing, duplicate rejection
```

3. Build the NIFTY 50 time series and features

This creates raw closes, aligned closes, feature rows, chronological splits, plots, and a manifest with hashes. The end date defaults to the latest completed date available to the script. Choose a new output folder for each snapshot.

```
phase1_args = ['--indices', 'NIFTY50', '--output', str(PHASE1_OUTPUT)]
if DATA_SOURCE == 'local_csv':
    phase1_args += ['--input-dir', str(LOCAL_CSV_DIR)]
elif DATA_SOURCE != 'yahoo':
    raise ValueError("DATA_SOURCE must be 'yahoo' or 'local_csv'")

phase1_nifty.main(phase1_args)
```

```
Downloading NIFTY50: ^NSEI

C:\Users\pqrth\OneDrive\Desktop\RegimeChanges-main\RegimeChanges-
main\.venv\Lib\site-packages\yfinance\utils.py:568: Pandas4Warning:
The inplace keyword in DataFrame.sort_index is deprecated and will be
removed in a future version (PDEP-8).
See
https://pandas.pydata.org/docs/dev/whatsnew/v3.1.0.html#deprecation-
inplace for more details.
  quotes.sort_index(inplace=True)
C:\Users\pqrth\OneDrive\Desktop\RegimeChanges-main\RegimeChanges-
main\.venv\Lib\site-packages\yfinance\multi.py:220: Pandas4Warning:
The inplace keyword in DataFrame.rename is deprecated and will be
```

```
removed in a future version (PDEP-8).
See
https://pandas.pydata.org/docs/dev/whatsnew/v3.1.0.html#deprecation-
inplace for more details.
  data.rename(columns=ctx.isins, inplace=True)
C:\Users\pqrth\OneDrive\Desktop\RegimeChanges-main\RegimeChanges-
main\.venv\Lib\site-packages\yfinance\multi.py:224: Pandas4Warning:
The inplace keyword in DataFrame.sort_index is deprecated and will be
removed in a future version (PDEP-8).
See
https://pandas.pydata.org/docs/dev/whatsnew/v3.1.0.html#deprecation-
inplace for more details.
  data.sort_index(level=0, axis=1, inplace=True)

Created C:\Users\pqrth\OneDrive\Desktop\RegimeChanges-main\
RegimeChanges-main\phase1_nifty50 | causal checks passed | inspect
manifest.json for coverage
```

4. Inspect the Phase 1 snapshot

Check that the manifest exists and that the feature file has complete rows before proceeding.

```
manifest_path = PHASE1_OUTPUT / 'manifest.json'
features_path = PHASE1_OUTPUT / 'features' / 'NIFTY50_features.csv'
manifest = json.loads(manifest_path.read_text(encoding='utf-8'))
features = pd.read_csv(features_path, parse_dates=['Date',
'source_date'])
print('Feature rows:', len(features))
print('Date range:', features['Date'].min(), 'to',
features['Date'].max())
print('Rows by split:')
display(features['split'].value_counts(dropna=False).rename_axis('spli
t').to_frame('rows'))
print('Complete rows:',
int(features['complete'].fillna(False).astype(bool).sum()))
print('Manifest source:', manifest.get('source'))
display(features.head())

Feature rows: 4130
Date range: 2010-01-04 00:00:00 to 2026-10-27 00:00:00
Rows by split:

            rows
split
train       2693
validation   741
test         696
```

```
Complete rows: 4069
Manifest source: Yahoo Finance via yfinance

        Date source_date  split  complete  log_return  abs_return
vol_20  \
0 2010-01-04         NaT  train     False         NaN         NaN
NaN
1 2010-01-05  2010-01-04  train     False         NaN         NaN
NaN
2 2010-01-06  2010-01-05  train     False    0.008696    0.008696
NaN
3 2010-01-07  2010-01-06  train     False    0.000739    0.000739
NaN
4 2010-01-08  2010-01-07  train     False   -0.003547    0.003547
NaN

   vol_60  mean_20  momentum_20  drawdown  corr_nifty_60
0     NaN      NaN          NaN       NaN            NaN
1     NaN      NaN          NaN   0.00000            NaN
2     NaN      NaN          NaN   0.00000            NaN
3     NaN      NaN          NaN   0.00000            NaN
4     NaN      NaN          NaN  -0.00354            NaN
```

5. Run Phase 2 checks

This verifies sequential detection behavior on synthetic series. It does not use your market data.

```
import phase2_cpd
phase2_cpd.main(['--self-test'])

PASS l2: synthetic break, prefix/future invariance, frozen
calibration, gap reset, terminal removal
PASS rbf: synthetic break, prefix/future invariance, frozen
calibration, gap reset, terminal removal
```

6. Run as-of change-point detection on validation

This uses the frozen PELT settings and excludes the test split. The decision log records when an alert could have been issued; the estimated break date is kept separately. Use a new, empty output directory for each experiment.

```
phase2_args = [
    '--input', str(PHASE1_OUTPUT),
    '--indices', 'NIFTY50',
    '--output', str(PHASE2_OUTPUT),
    '--window', '252',
    '--min-size', '20',
    '--jump', '5',
```

```
    '--every', '5',
    '--confirmations', '2',
    '--tolerance', '5',
    '--max-age', '60',
    '--cooldown', '20',
    '--penalty-l2', '10.0',
    '--penalty-rbf', '5.0',
]
phase2_cpd.main(phase2_args)
```

```
NIFTY50 / l2: sequential detection (3434 feature rows)
  saved 741 decisions, 6 alerts
NIFTY50 / rbf: sequential detection (3434 feature rows)
  saved 741 decisions, 9 alerts
Created C:\Users\pqrth\OneDrive\Desktop\RegimeChanges-main\
RegimeChanges-main\phase2_nifty50_validation | test excluded
```

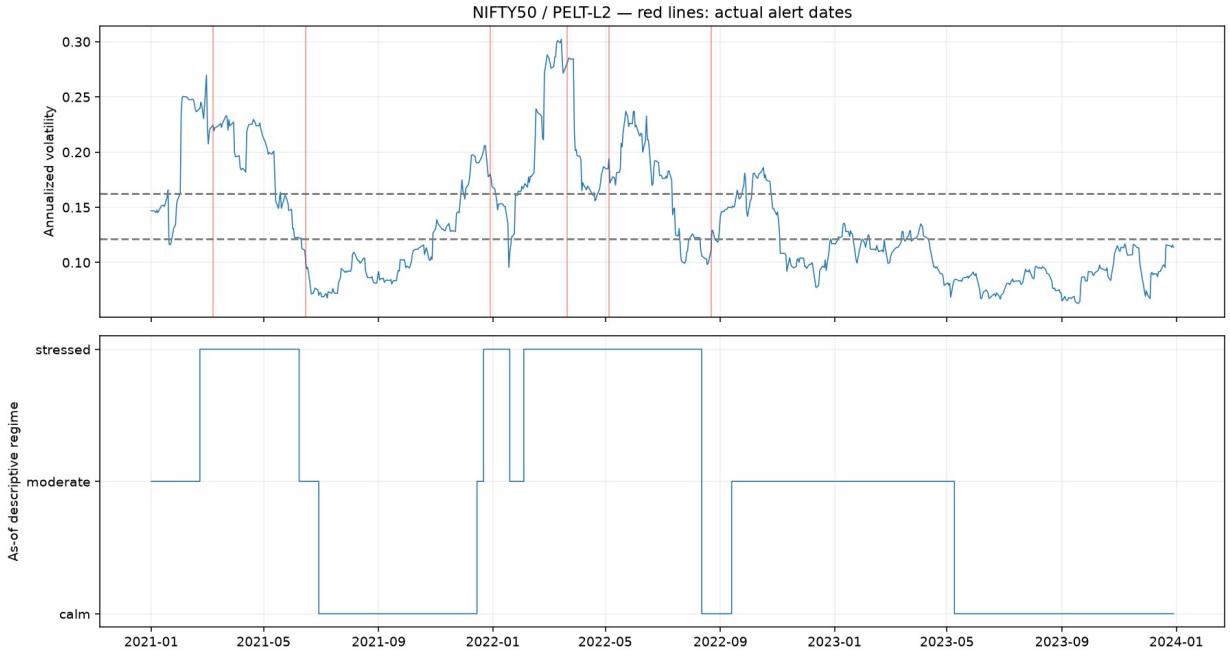
7. Review Phase 2 validation outputs

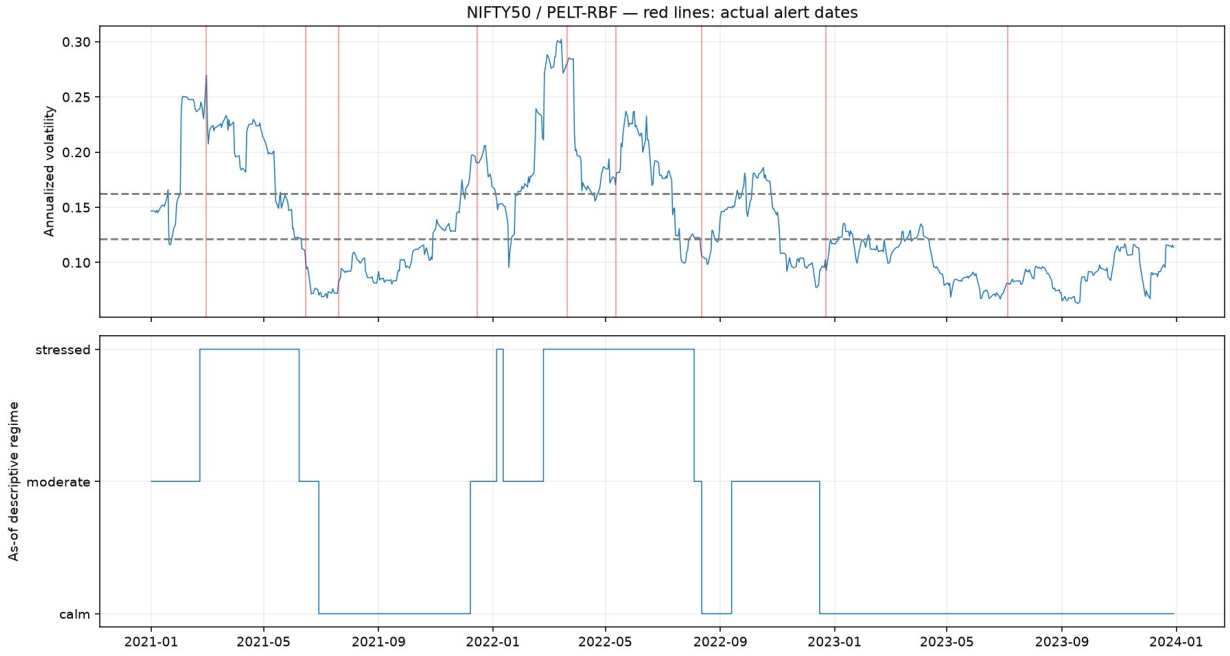
The output folder contains decision logs, confirmed event logs, training-only calibration details, validation timeline plots, a summary, and a run manifest. Verify `test_enabled` is false and the manifest reports window 252 and min_size 20. Do not use test results to tune parameters.

```
summary_path = PHASE2_OUTPUT / 'summary.csv'
summary = pd.read_csv(summary_path)
display(summary)
print('Output files:')
for path in sorted(PHASE2_OUTPUT.iterdir()):
    print(path.name)

from IPython.display import Image, display
for model in ('l2', 'rbf'):
    display(Image(filename=str(PHASE2_OUTPUT /
f'NIFTY50_{model}_timeline.png')))
```

```
     index model       split  rows  checks  alerts  unknown_rows
calm_rows  \
0  NIFTY50    l2  validation   741     149       6             0
296
1  NIFTY50   rbf  validation   741     149       9             0
386

   moderate_rows  stressed_rows
0            225            220
1            170            185

Output files:
NIFTY50_l2_calibration.json
NIFTY50_l2_decisions.csv
```

```
NIFTY50_l2_events.csv
NIFTY50_l2_timeline.png
NIFTY50_rbf_calibration.json
NIFTY50_rbf_decisions.csv
NIFTY50_rbf_events.csv
NIFTY50_rbf_timeline.png
run_manifest.json
summary.csv
```





8. Phase 3: fit HMM on training and compare on validation

This fits a three-state diagonal Gaussian HMM on training rows only, using the Phase 1 training-only scaler and `log_return` and `vol_20`. Ten seeded initializations are compared using training likelihood only. Validation labels use recursive forward filtering with the training filter state carried forward; there is no backward smoothing or test use. The script also verifies that the Phase 2 baseline used the frozen 252/20 configuration and excluded test data.

Use a new, empty `PHASE3_HMM_OUTPUT`. The HMM is a validation benchmark only; it is not applied to the held-out test period.

```
import phase3_hmm

phase3_hmm.main([
    '--input', str(PHASE1_OUTPUT),
    '--phase2', str(PHASE2_OUTPUT),
    '--output', str(PHASE3_HMM_OUTPUT),
    '--states', '3',
    '--restarts', '10',
    '--seed', '1701',
])
```

```
Created C:\Users\pqrth\OneDrive\Desktop\RegimeChanges-main\
RegimeChanges-main\phase3_hmm_validation | HMM fit on training only |
forward-filtered validation | test excluded
{'comparison': 'HMM vs PELT-L2', 'validation_dates': 741,
'label_agreement': 0.601889338731444, 'cohen_kappa':
0.38239008637130334, 'adjusted_rand_index': 0.283599870170364,
'pelt_confirmed_alerts': 6,
'pelt_estimated_breaks_with_hmm_transition_within_5_sessions': '4/6',
'median_nearest_hmm_transition_gap_sessions': 3.0}
{'comparison': 'HMM vs PELT-RBF', 'validation_dates': 741,
'label_agreement': 0.7273954116059379, 'cohen_kappa':
0.5503789346574268, 'adjusted_rand_index': 0.4642031024247661,
'pelt_confirmed_alerts': 9,
'pelt_estimated_breaks_with_hmm_transition_within_5_sessions': '7/9',
'median_nearest_hmm_transition_gap_sessions': 3.0}
```

Review HMM comparison and validation timelines

Agreement and transition proximity compare unsupervised methods; they are not accuracy against known regimes. Review the table and chart together.

```
hmm_comparison = pd.read_csv(PHASE3_HMM_OUTPUT /
'hmm_vs_pelt_validation.csv')
display(hmm_comparison)

hmm_validation = pd.read_csv(
    PHASE3_HMM_OUTPUT / 'NIFTY50_hmm_validation_probabilities.csv',
```

```
    parse_dates=['Date'],
)
label_order = {'calm': 0, 'moderate': 1, 'stressed': 2}
fig, axes = plt.subplots(3, 1, figsize=(13, 7), sharex=True)
for label in ('calm', 'moderate', 'stressed'):
    axes[0].plot(hmm_validation['Date'],
hmm_validation[f'prob_{label}'], label=label)
axes[0].set_ylabel('HMM filtered probability')
axes[0].legend(ncol=3)
for ax, model in zip(axes[1:], ('l2', 'rbf')):
    pelt = pd.read_csv(PHASE2_OUTPUT /
f'NIFTY50_{model}_decisions.csv', parse_dates=['Date'])
    joined = hmm_validation[['Date', 'hmm_regime']].merge(
        pelt[['Date', 'regime']], on='Date', how='inner',
validate='one_to_one'
    )
    ax.step(joined['Date'], joined['hmm_regime'].map(label_order),
where='post', label='HMM')
    ax.step(joined['Date'], joined['regime'].map(label_order),
where='post', label=f'PELT-{model.upper()}')
    ax.set_yticks([0, 1, 2], ['calm', 'moderate', 'stressed'])
    ax.set_ylabel('Regime')
    ax.legend(ncol=2)
for ax in axes:
    ax.grid(alpha=.2)
axes[-1].set_xlabel('Validation date')
fig.tight_layout()
plt.show()
```

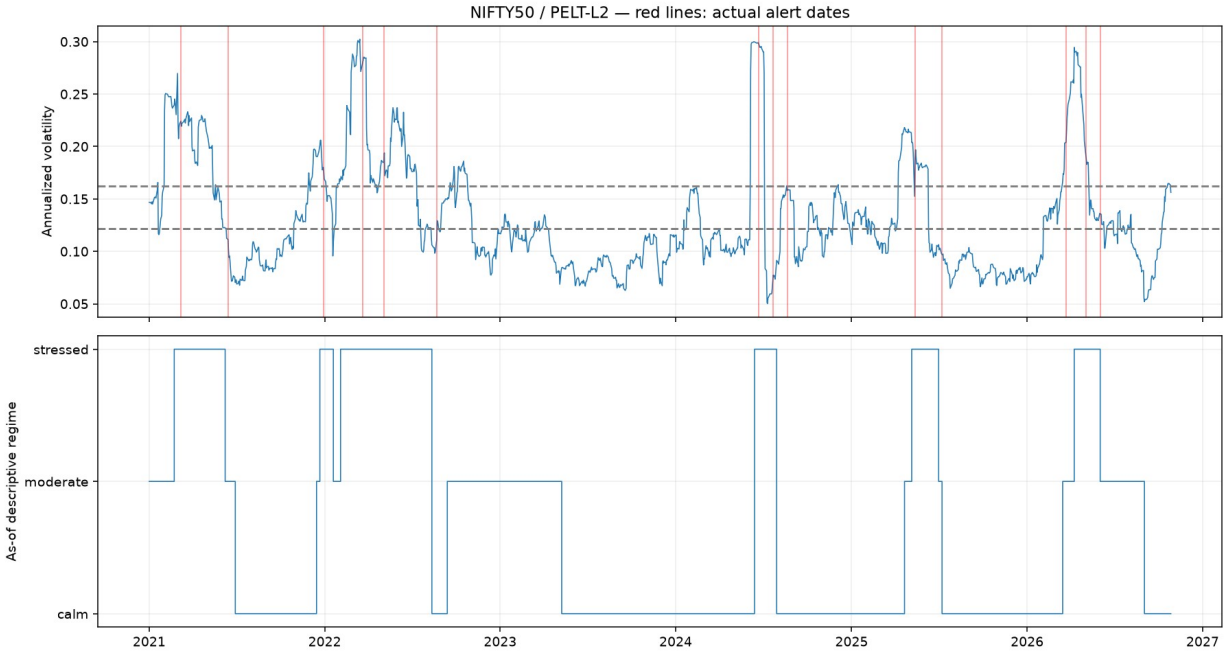
```
        comparison  validation_dates  label_agreement  cohen_kappa  \
0   HMM vs PELT-L2               741         0.601889     0.382390
1  HMM vs PELT-RBF               741         0.727395     0.550379

   adjusted_rand_index  pelt_confirmed_alerts  \
0             0.283600                      6
1             0.464203                      9

  pelt_estimated_breaks_with_hmm_transition_within_5_sessions  \
0                                                        4/6
1                                                        7/9

   median_nearest_hmm_transition_gap_sessions
0                                         3.0
1                                         3.0

C:\Users\pqrth\AppData\Local\Temp\ipykernel_32548\1741462526.py:28:
UserWarning: FigureCanvasAgg is non-interactive, and thus cannot be
shown
  plt.show()
```

9. Optional final test: open the held-out period once

The test run for the recorded Phase 1 snapshot has already been completed. Leave `RUN_FINAL_TEST = False` when reviewing or re-running validation. For a new, deliberate final evaluation on a new frozen snapshot, first confirm all design choices are fixed, then set `RUN_FINAL_TEST = True`, use a new empty `FINAL_OUTPUT`, and execute this cell once. Do not use test results to tune the detector.

```
if RUN_FINAL_TEST:
    phase2_cpd.main([
        '--input', str(PHASE1_OUTPUT),
        '--indices', 'NIFTY50',
        '--output', str(FINAL_OUTPUT),
        '--window', '252',
        '--min-size', '20',
        '--jump', '5',
        '--every', '5',
        '--confirmations', '2',
        '--tolerance', '5',
        '--max-age', '60',
        '--cooldown', '20',
        '--penalty-l2', '10.0',
        '--penalty-rbf', '5.0',
        '--include-test',
    ])
    final_summary = pd.read_csv(FINAL_OUTPUT / 'summary.csv')
    display(final_summary)
    from IPython.display import Image, display
    for model in ('l2', 'rbf'):
        display(Image(filename=str(FINAL_OUTPUT /
f'NIFTY50_{model}_timeline.png')))
else:
    print('Held-out test remains closed. Set RUN_FINAL_TEST=True only
for a deliberate one-time final run.')
```

```
NIFTY50 / l2: sequential detection (4130 feature rows)
  saved 1437 decisions, 14 alerts
NIFTY50 / rbf: sequential detection (4130 feature rows)
  saved 1437 decisions, 18 alerts
Created C:\Users\pqrth\OneDrive\Desktop\RegimeChanges-main\
RegimeChanges-main\phase2_nifty50_final_test | test included
```

```
     index model       split   rows   checks   alerts   unknown_rows
calm_rows  \
0  NIFTY50    l2  validation    741      149        6              0
296
1  NIFTY50    l2        test    696      139        8              0
496
2  NIFTY50   rbf  validation    741      149        9              0
```

```
386
3  NIFTY50    rbf         test    696      139        9               0
376

   moderate_rows  stressed_rows
0            225            220
1             95            105
2            170            185
3            215            105
```